

PRIVATE CREDIT

Early-Warning & Loss Monitor

8-Week Build Plan

A practical roadmap to a paid pilot for a focused B2B credit-risk product.

Product proposition

Upload a loan portfolio → identify deteriorating borrowers → quantify the impact on PD, LGD and Expected Loss → explain why.

Prepared as a founder build plan | August 2026

Executive Summary

The aim of the first eight weeks is not to build the most sophisticated credit model. It is to reach a paid design-partner pilot with a narrowly useful product: a credit analyst in software for private-credit teams.

At the end of Week 8, a credit manager should be able to upload an Excel or CSV loan book, see which borrowers are deteriorating, understand the changes in PD/LGD/Expected Loss (EL), and run an immediate stress scenario.

Satellite data, NLP and complex ML are deliberately deferred. The first product should be explainable, auditable and useful in a ten-minute demo.

Week-8 customer experience

PORTFOLIO EARLY-WARNING DASHBOARD

Portfolio EAD	£487m
Expected Loss	£14.3m
Change since previous period	+£2.1m

Highest-risk borrowers

	EAD	PD	LGD	EL	Score	
ABC Logistics	£18m	12.1%	51%	£1.11m	87	CRITICAL
XYZ Foods	£24m	7.4%	43%	£0.76m	72	RED
Delta Mfg	£11m	5.8%	39%	£0.25m	61	RED

ABC Logistics | EBITDA -24% | Debt/EBITDA 4.1 -> 5.3x
Interest cover 2.4 -> 1.6x | Covenant headroom 7%
PD 6.0% -> 12.1% | LGD 45% -> 51% | EL £486k -> £1.11m

The main demo claim is not a black-box prediction. It is a clear, reproducible explanation of the changes that matter.

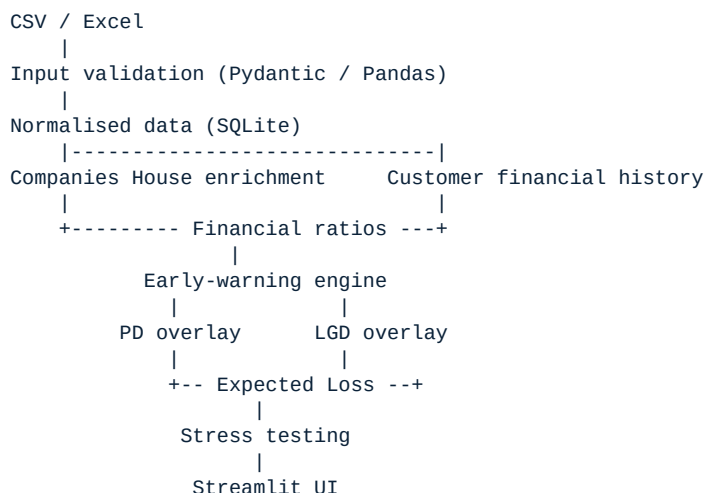
Core equation

$$EL = EAD \times PD \times LGD$$

Track both the amount and the movement: $\Delta EL = EL(t) - EL(t-1)$.

Simple MVP Architecture

Do not build a complicated SaaS architecture initially. Use **Python + Pandas + SQLite + Streamlit + Pytest**. Avoid React, Kubernetes, microservices and a separate API in the first sprint.



Build choices

- Streamlit is appropriate for a commercial prototype; FastAPI can be introduced later if the application/API layer needs separating.
- Use Arelle for XBRL/iXBRL processing through its Python API.
- Store provenance for every financial input: customer upload, Companies House, calculated, or manual override.

Week 1 - Build the credit calculation engine

Goal: forget the UI initially. Build a Python package that converts borrower data into ratios and expected loss.

```
borrower, EAD, PD, LGD, revenue, EBITDA, debt, cash, interest
ABC Ltd, 18000000, 0.06, 0.45, 50000000, 8000000, 35000000, 4000000, 5000000
```

```
{
  "leverage": 4.375,
  "interest_cover": 1.6,
  "net_debt": 31_000_000,
  "expected_loss": 486_000
}
```

Minimum calculations

Metric	Formula
Net debt	Debt - Cash
Leverage	Debt / EBITDA
Net leverage	(Debt - Cash) / EBITDA
Interest cover	EBITDA / Interest Expense
EBITDA margin	EBITDA / Revenue
Expected loss	EAD x PD x LGD

Also calculate period-on-period changes in EBITDA, Revenue, Leverage and Interest Cover.

Core data model

borrower: borrower_id, company_name, company_number, sector, country
exposure: loan_id, borrower_id, ead, currency, seniority, maturity, baseline_pd, baseline_lgd
financials: borrower_id, period, revenue, ebitda, debt, cash, interest_expense, assets
covenants: loan_id, max_leverage, min_interest_cover, min_liquidity
collateral: loan_id, collateral_type, value, valuation_date
risk results: loan_id, date, risk score, pd, lqd, expected_loss, risk band

Definition of done: run `analyse_portfolio("portfolio.csv")` on a 50-borrower synthetic portfolio and return a DataFrame ranked by expected-loss deterioration; calculations have unit tests.

Week 2 - Build the Early-Warning Model

This is the most important week. Do **not** train an ML model yet. Build an explainable, capped-at-100 scorecard.

Signal	Condition	Risk points
EBITDA	decline >20%	+15
EBITDA	decline 10-20%	+8
Leverage	increase >1x	+15
Leverage	above covenant	+20
Interest cover	<1.5x	+15
Interest cover	<1.0x	+20
Covenant headroom	<10%	+15
Covenant	breach	+25
Cash	decline >25%	+10
Revenue	decline >15%	+8

Risk bands

Score	Band
0-24	GREEN
25-49	AMBER
50-74	RED
75-100	CRITICAL

These thresholds are product assumptions, not statistically calibrated probabilities. Say this explicitly throughout the prototype.

PD overlay

Turn a lender-supplied baseline PD into an **indicative stressed PD** using risk-band multipliers: Green 1.0x; Amber 1.5x; Red 2.5x; Critical 4.0x.

$$\lambda = -\ln(1 - PD_0) \quad | \quad \lambda^* = m \times \lambda \quad | \quad PD^* = 1 - e^{(-\lambda^*)}$$

This hazard-rate transformation behaves better as PD rises than multiplying PD directly. Never present it as a validated regulatory PD model.

Week 3 - Add Companies House

The demo becomes much more compelling when a company number can retrieve basic public-company information and electronically filed accounts.

```
get_company("01234567")
```

```
# {"company_name": "...", "sic": "...", "company_status": "active",  
#  "accounts_due": "..."}  
  
extract_financials(xbrl)
```

Initially map only: Revenue, Operating Profit, Profit Before Tax, Cash, Current Assets, Current Liabilities, Total Assets, Creditors, Equity and Employees. Do not attempt to interpret every Companies House taxonomy tag.

EBITDA: value = £7.4m | source = CUSTOMER_UPLOAD

Cash: value = £2.1m | source = COMPANIES_HOUSE

Current liabilities: value = £8.9m | source = COMPANIES_HOUSE

Target: get 50 real UK companies flowing through the pipeline. Missing-data handling is part of the product. Provenance is not optional; institutional users care about lineage.

Week 4 - Build the Dashboard

Build only four Streamlit screens. Keep every ranking sortable.

Screen	What it must show
1. Portfolio	EAD, current EL, stressed EL, red and critical borrower counts; sortable ranking with Borrower, EAD, Score, PD, LGD, EL and Delta EL.
2. Borrower	Risk score, PD, LGD, EL and trend charts for Revenue, EBITDA, Leverage, Interest Cover, Cash, PD and EL.
3. Risk Drivers	A point-by-point explanation: leverage covenant breached +20; EBITDA declined 27% +15; interest cover below 1.5x +15; cash declined 31% +10; revenue declined 17% +8.
4. Stress Test	Sliders for EBITDA, Revenue, Collateral and interest-rate shocks; immediate portfolio EL recalculation.

Avoid: “AI predicts ABC is high risk.” Credit professionals will reasonably ask why. The Risk Drivers screen is essential.

Week 5 - Expected Loss Attribution

Make attribution the differentiator. Do not merely report an EL movement; explain it. For $EL = EAD \times PD \times LGD$, decompose Delta EL into EAD, PD and LGD contributions using **Shapley attribution** (average marginal effects across possible calculation orders).

March EL	£486k	
June EL	£1.11m	
Increase	£624k	
PD deterioration	+£402k	64%
LGD deterioration	+£151k	24%
Exposure increase	+£71k	12%

Primary PD signals: Leverage 35% | EBITDA 29% | Interest cover 21% | Liquidity 15%

This is the sort of feature a credit committee could find genuinely useful.

Week 6 - Validation and Auditability

Do not add flashy features. Try to break the product.

Test case	Expected result
EBITDA falls; everything else constant	Risk up; PD up; EL up
Debt falls substantially	Leverage down; risk down
Collateral collapses	PD unchanged; LGD up; EL up
EAD increases	PD unchanged; LGD unchanged; EL up

Calculation date: 2026-08-23
Input PD 6.00% | Risk multiplier 2.00 | Calculated PD 11.64%
Reason: risk band = RED
Risk score: Leverage +20 | EBITDA +15 | Interest cover +15 | Liquidity +10
Total: 60

A user should be able to reproduce every calculation. That may be a competitive advantage over generic “AI credit scoring”. Build towards hundreds of automated tests.

Week 7 - Customer Discovery and Demos

Stop coding for part of the week and start selling. The initial target is UK private-credit/direct-lending teams managing roughly £100m-£5bn, rather than banks or large asset managers with lengthy procurement.

Prioritise specialist direct lenders, private debt funds, asset managers, credit hedge funds, family offices, specialty-finance companies and small banks. Find around 50 people titled Head of Credit, Private Credit Portfolio Manager, Credit Risk Director, Head of Portfolio Monitoring or Investment Director - Private Debt.

Discovery message

"I've built a prototype that takes a private-credit loan book and ranks borrowers by deterioration risk using leverage, interest coverage, liquidity and covenant headroom. It also shows how those changes affect portfolio expected loss. I'm speaking with private-credit teams to understand how they currently monitor deterioration between reporting dates. Would you be willing to spend 20 minutes walking me through your current process?"

Do not lead with "innovative AI-powered platform." The objective is discovery, not a polished sales pitch.

Ten-minute demo flow

Minute	Action
1	Upload portfolio.
2-3	Show the ten borrowers requiring attention.
4-5	Open borrower #1 and show exactly why it was flagged.
6-7	Show PD, LGD and EL movement.
8	Run a -25% EBITDA scenario.
9	Show the portfolio EL increase.
10	Ask: "How do you currently do this?" Then stop talking.

Week 8 - Convert a User into a Pilot

Do not obsess about subscription pricing. Offer a 4-6 week design-partner pilot and test a first-customer range of **£1,500-£5,000**. This is a range to test, not a claim about market-standard pricing.

Pilot includes
Portfolio upload
Automated financial ratios
Borrower risk ranking
PD/LGD overlays
Expected-loss calculations
Covenant monitoring

Pilot includes
Scenario analysis
Weekly risk report

Do not promise satellite monitoring in the pilot. The aim is to learn whether somebody will pay, not whether they like the demo.

Week-8 sales target

20 conversations -> 5 demos -> 2 serious pilot discussions -> 1 paid pilot

If 20 conversations lead to zero pilot interest, do not spend another six months coding. Diagnose the proposition. If teams repeatedly say covenant extraction is the real pain, that becomes the product. Customer contact begins before the system is perfect.

Scope Discipline and Product Evolution

What deliberately does **not** belong in the first eight weeks: LLM chatbot, satellite change detection, sophisticated ML, multi-factor Monte Carlo models, React frontend, mobile app, Kubernetes, multi-tenancy, automated underwriting, portfolio optimisation, ESG scoring, and dozens of data APIs.

Every one can consume weeks without establishing whether someone will pay.

Satellite imagery: Phase II

Once the financial-monitoring core is valuable, physical activity signals can become a differentiator: financial monitoring + covenant monitoring + Companies House + satellite activity signal.

ABC Distribution Ltd
Financial warning: EBITDA -19%
Physical activity warning: warehouse activity proxy -27%
Credit deterioration score: 84

Copernicus Sentinel-2 makes a potential later layer attractive because its data are available free of charge to commercial users. It offers four 10m bands, six 20m bands and three 60m bands. Do not put this into the first sprint.

Suggested repository layout

```
credit-monitor/  
■■■ app/          streamlit_app.py, portfolio.py, borrower.py, stress.py  
■■■ credit/       ratios.py, expected_loss.py, pd_overlay.py, lgd_overlay.py,  
■■              early_warning.py, attribution.py, scenarios.py  
■■■ data/         companies_house.py, xbrl.py, validation.py  
■■■ database/     models.py, database.py  
■■■ tests/        test_ratios.py, test_pd.py, test_lgd.py, test_el.py, test_scenarios.py  
■■■ sample_data/ demo_portfolio.csv  
■■■ requirements.txt  
■■■ README.md
```

The actual MVP workflow

```
Upload portfolio  
-> 487 loans loaded; 3 validation errors  
Analyse  
-> 17 borrowers deteriorated; 6 require immediate review  
Open ABC Ltd  
-> Risk score 82; leverage +1.3x; EBITDA -24%; ICR below 1.5x; headroom 6%  
-> Baseline EL £486k; Current EL £1.11m; Increase £624k  
Stress  
-> EBITDA -20%; Interest rates +200bp; Portfolio EL £14.3m -> £19.8m
```

That is a product.

Success Metric, Day 1 and Sources

The most important metric is not model accuracy yet. Track the number of users who ask: “Can I try this on our portfolio?”

“That’s interesting” is weak. “Can I send you a portfolio?” is strong. “How much does it cost?” is stronger. “Can you sign an NDA?” is very strong. A paid pilot is the real validation.

If starting tomorrow

Create a 30-borrower synthetic private-credit portfolio with two reporting periods, EAD, PD, LGD, revenue, EBITDA, debt, cash, interest expense and three covenants. Then write these five functions:

```
calculate_ratios()  
calculate_expected_loss()  
calculate_risk_score()  
calculate_stressed_pd()  
rank_borrowers()
```

Once those work, you have the nucleus of the business.

Sources and implementation references

Companies House: Companies House public company information and services; its data-products guidance describes free bulk data, including accounts data.

<https://www.gov.uk/government/organisations/companies-house/about-our-services>

<https://www.gov.uk/guidance/companies-house-data-products>

Arele: open-source XBRL platform with Python support.

<https://arelle.org/arelle/>

Streamlit: implementation documentation and licence information.

<https://docs.streamlit.io/develop/concepts/configuration/overview>

<https://github.com/streamlit/streamlit/blob/develop/LICENSE>

Copernicus: data-access information, including Sentinel programme data availability.

<https://www.copernicus.eu/en/access-data>

Important note: scorecard thresholds, PD multipliers and pilot pricing are deliberate prototype assumptions to test with users, not regulatory-model calibrations, market-price claims or investment advice.